Product Summary

$V_{(BR)DSS}$	$R_{DS(on)\ max}$	I_D T_A = +25°C (Note 6)
-100V	150mΩ@ V_{GS} = -10V	-5.9A
	190mΩ@ V_{GS} =-6V	-5.1A

Description

This new generation trench MOSFET from Zetex features a unique structure combining the benefits of low on-resistance and fast switching, making it ideal for high efficiency power management applications.

Applications

- DC-DC Converters
- Power Management Functions
- Disconnect Switches
- Motor Control

Features and Benefits

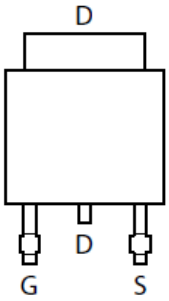
- Low On-Resistance
- Fast Switching Speed
- Low Threshold
- Low Gate Drive
- DPAK Package
- **Totally Lead-Free & Fully RoHS Compliant (Notes 1 & 2)**
- **Halogen and Antimony Free. "Green" Device (Note 3)**
- **Qualified to AEC-Q101 Standards for High Reliability**

Mechanical Data

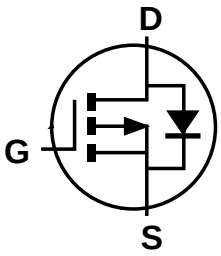
- Case: TO252 (DPAK)
- Case Material: Molded Plastic, "Green" Molding Compound. UL Flammability Classification Rating 94V-0 (Note 1)
- Moisture Sensitivity: Level 1 per J-STD-020
- Terminals Connections: See diagram below
- Terminals: Finish - Matte Tin Annealed over Copper Leadframe. Solderable per MIL-STD-202, Method 208 e3
- Weight: 0.315 grams (Approximate)

TO252

Top View



Top View
Pin Out



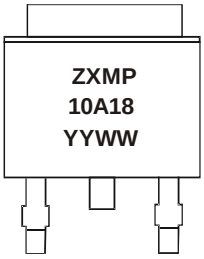
Equivalent circuit

Ordering Information (Note 4)

Part Number	Compliance	Case	Packaging
ZXMP10A18KTC	Standard	TO252	2,500/Tape & Reel

Notes:
1. No purposely added lead. Fully EU Directive 2002/95/EC (RoHS) & 2011/65/EU (RoHS 2) compliant.
2. See http://www.diodes.com/quality/lead_free.html com for more information about Diodes Incorporated's definitions of Halogen- and Antimony-free, "Green" and Lead-free.
3. Halogen- and Antimony-free "Green" products are defined as those which contain <900ppm bromine, <900ppm chlorine (<1500ppm total Br + Cl) and <1000ppm antimony compounds.
4. For packaging details, go to our website at http://www.diodes.com/products/packages.html.

Marking Information



ZXMP10A18 = Product Type Marking Code
YYWW = Date Code Marking
YY = Year (ex: 10 = 2010)
WW = Week (01 - 53)

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Maximum Ratings (@T_A = +25°C, unless otherwise specified.)

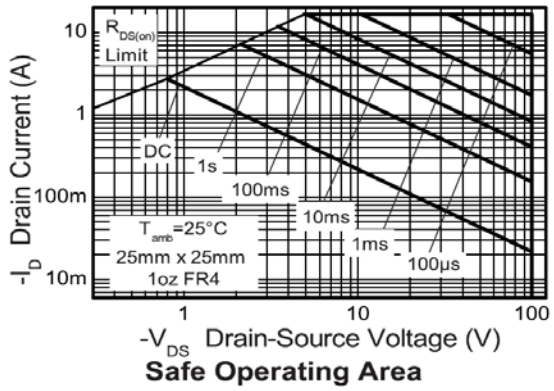
Characteristic		Symbol	Value	Units
Drain-Source Voltage		V_{DSS}	-100	V
Gate-Source Voltage		V_{GSS}	±20	V
Continuous Drain Current	T_A = +25°C (Note 6)	I_D	-5.9	A
	T_A = +70°C (Note 6)		-4.7	
	T_A = +25°C (Note 5)		-3.8	
Pulsed Drain Current (Note 7)		I_{DM}	-21.1	A
Continuous Source Current (Body Diode) (Note 6)		I_S	-10	A
Pulsed Source Current (Body Diode) (Note 7)		I_{SM}	-21.1	A

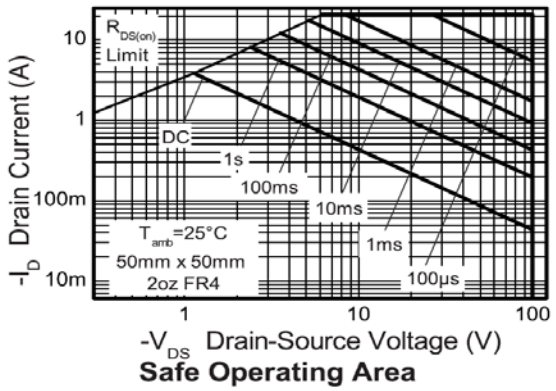
Thermal Characteristics (@T_A = +25°C, unless otherwise specified.)

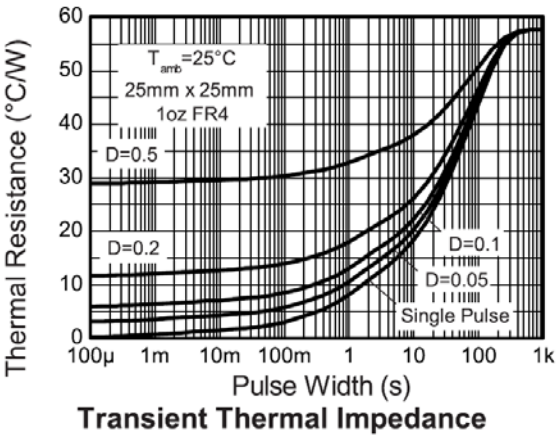
Characteristic		Symbol	Value	Units
Total Power Dissipation (Note 5) Linear Derating Factor	T_A = +25°C (Note 5)	P_D	4.3	W
			34.4	mW/°C
	T_A = +25°C (Note 6)		10.2	W
			81.3	mW/°C
	T_A = +25°C (Note 8)		2.17	W
			17.4	mW/°C
Thermal Resistance, Junction to Ambient	(Note 5)	$R_{\theta JA}$	29	°C/W
	(Note 6)		12.3	
	(Note 8)		57.6	
Operating and Storage Temperature Range		T_J, T_{STG}	-55 to +150	°C

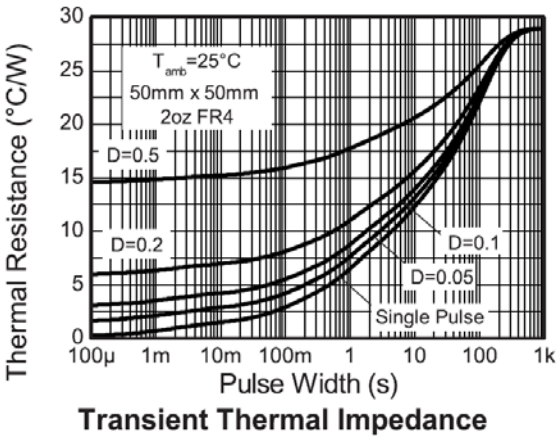
Notes:
5. For a device surface mounted on 50mm x 50mm x 1.6mm FR4 PCB with high coverage of single sided 2oz copper, in still air conditions.
6. For a device surface mounted on FR4 PCB measured at t ≤10 sec.
7. Repetitive rating on 50mm x 50mm x 1.6mm FR4 PCB, D=0.02, pulse width=300µs – pulse width limited by maximum junction temperature.
8. For a device surface mounted on 25mm x 25mm x 1.6mm FR4 PCB with high coverage of single sided 1oz copper, in still air conditions.

Thermal characteristics









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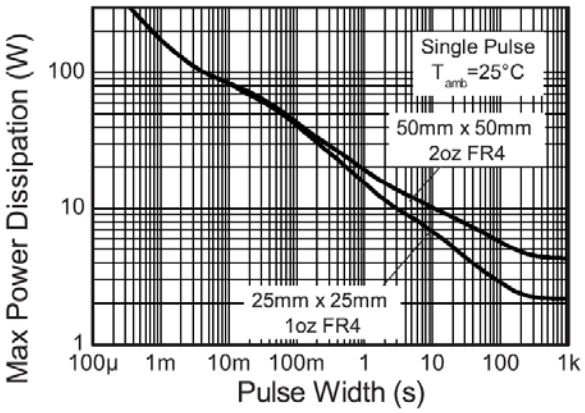
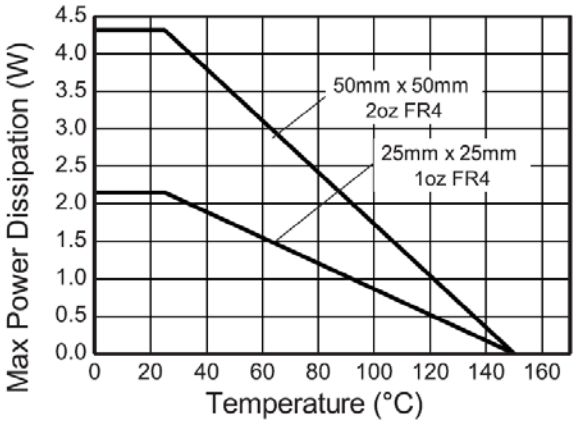


Pulse Power Dissipation



Derating Curve

Electrical Characteristics (@T_A = +25°C, unless otherwise specified.)

Characteristic	Symbol	Min	Typ	Max	Unit	Test Condition
OFF CHARACTERISTICS						
Drain-Source Breakdown Voltage	BV_{DSS}	-100	—	—	V	V_{GS} = 0V, I_D = -250µA
Zero Gate Voltage Drain Current	I_{DSS}	—	—	-1	µA	V_{DS} = -100V, V_{GS} = 0V
Gate-Source Leakage	I_{GSS}	—	—	±100	nA	V_{GS} = ±20V, V_{DS} = 0V
ON CHARACTERISTICS						
Gate Threshold Voltage	$V_{GS(th)}$	-2	—	-4	V	V_{DS} = V_{GS}, I_D = -250µA
Static Drain-Source On-Resistance (Note 9)	$R_{DS\ (ON)}$	—	—	150 190	mΩ	V_{GS} = -10V, I_D = -2.8A V_{GS} = -6V, I_D = -2.4A
Forward Transconductance (Notes 9 & 11)	g_{fs}	—	6	—	S	V_{DS} = -15V, I_D = -2.8A
DYNAMIC CHARACTERISTICS (Note 11)						
Input Capacitance	C_{iss}	—	1055	—	pF	V_{DS} = -50V, V_{GS} = 0V, f = 1MHz
Output Capacitance	C_{oss}	—	90	—	pF	
Reverse Transfer Capacitance	C_{rss}	—	76	—	pF	
SWITCHING CHARACTERISTICS (Notes 10 & 11)						
Turn-On Delay Time	$t_{d(on)}$	—	4.9	—	ns	V_{DS} = -50V, V_{GS} = -10V, I_D = -1A, R_G = 6Ω
Rise Time	t_r	—	6.8	—		
Turn-On Delay Time	$t_{d(off)}$	—	33.9	—		
Rise Time	t_f	—	17.9	—		
Total Gate Charge	Q_g	—	26.9	—	nC	V_{DS} = -50V, V_{GS} = -10V, I_D = -2.8A
Gate-Source Charge	Q_{gs}	—	3.9	—		
Gate-Drain Charge	Q_{gd}	—	10.2	—		
SOURCE-DRAIN DIODE CHARACTERISTICS						
Diode Forward Voltage (Note 9)	V_{SD}	—	-0.85	-0.95	V	T_J = +25°C , V_{GS} = 0V, I_S = -3.5A
Reverse Recovery Time (Note 11)	t_{rr}	—	49	—	ns	T_J = +25°C, I_S = -2.8A, di/dt=100A/µs,
Reverse Recovery Charge (Note 11)	Q_{rr}	—	107	—	nC	

Notes: 9. Measured under pulsed conditions. Pulse width ≤ 300µs; duty cycle ≤ 2%.
10. Switching characteristics are independent of operating junction temperature.
11. For design aid only, not subject to production testing.

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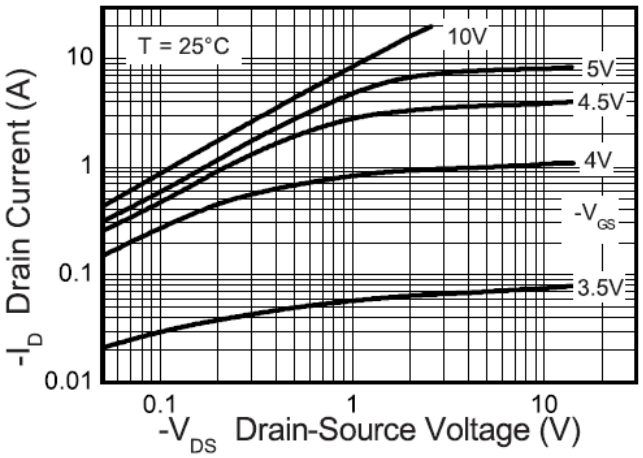
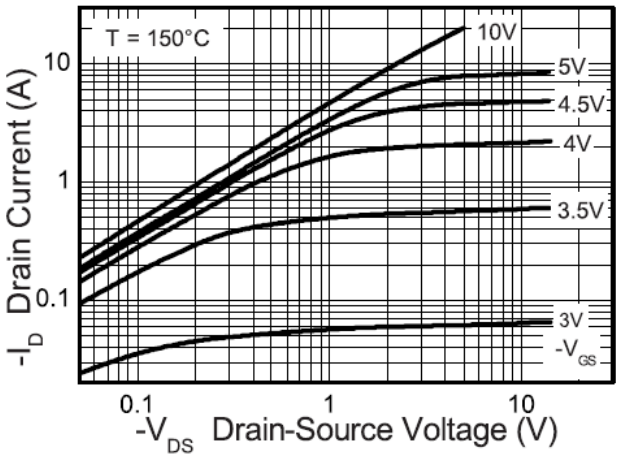
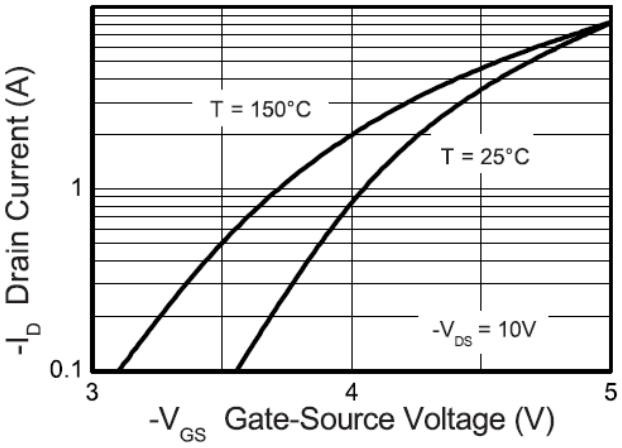
Typical characteristics



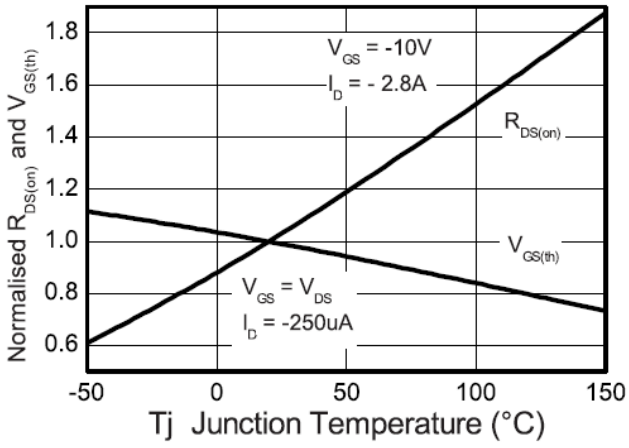
Output Characteristics



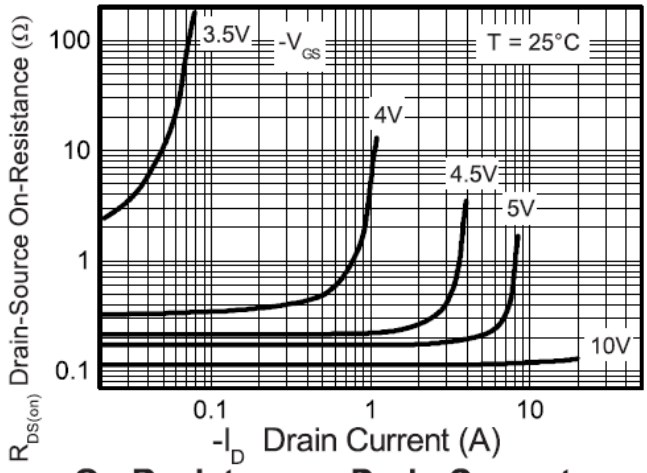
Output Characteristics



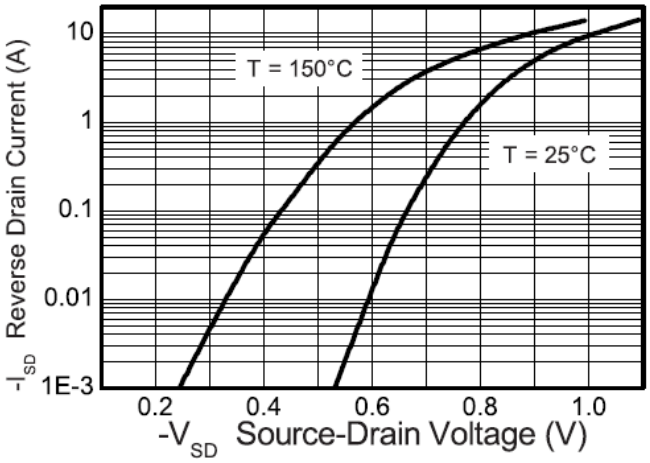
Typical Transfer Characteristics



Normalised Curves v Temperature



On-Resistance v Drain Current



Source-Drain Diode Forward Voltage

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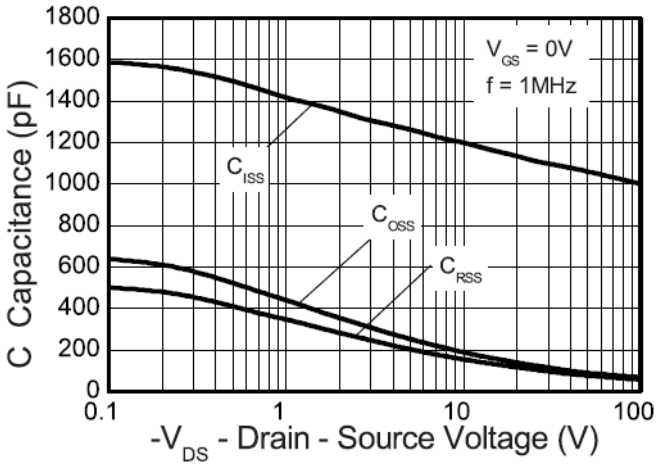
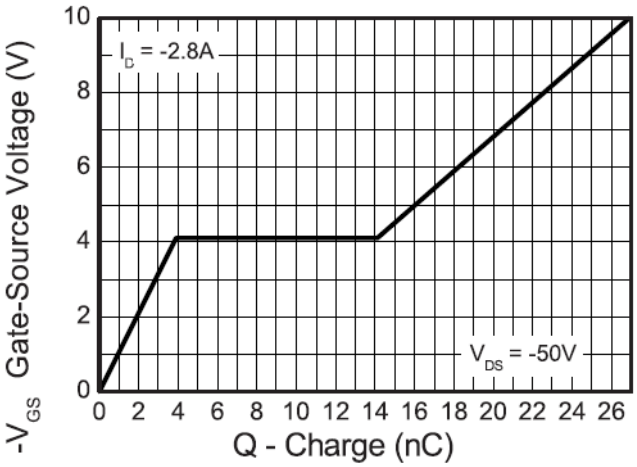
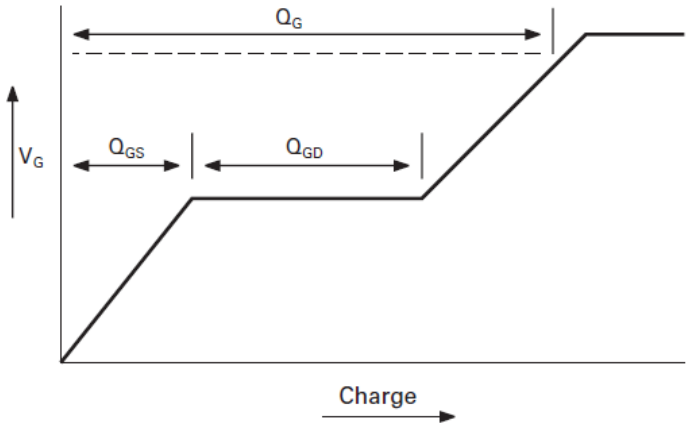
Typical characteristics



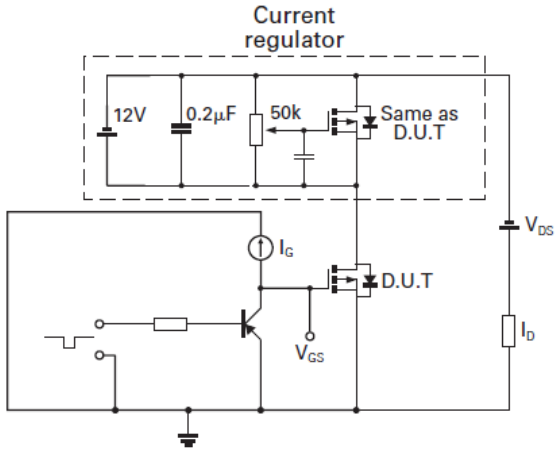
Capacitance v Drain-Source Voltage



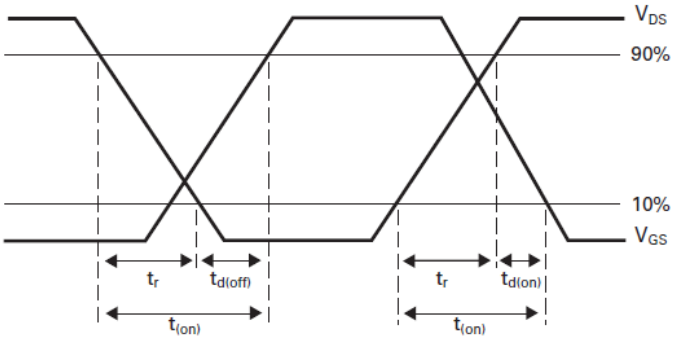
Gate-Source Voltage v Gate Charge



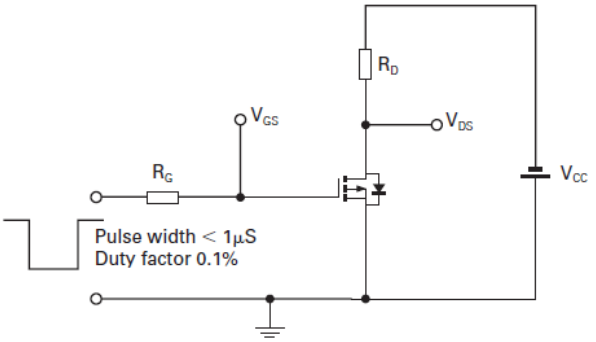
Basic gate charge waveform



Gate charge test circuit



Switching time waveforms



Switching time test circuit

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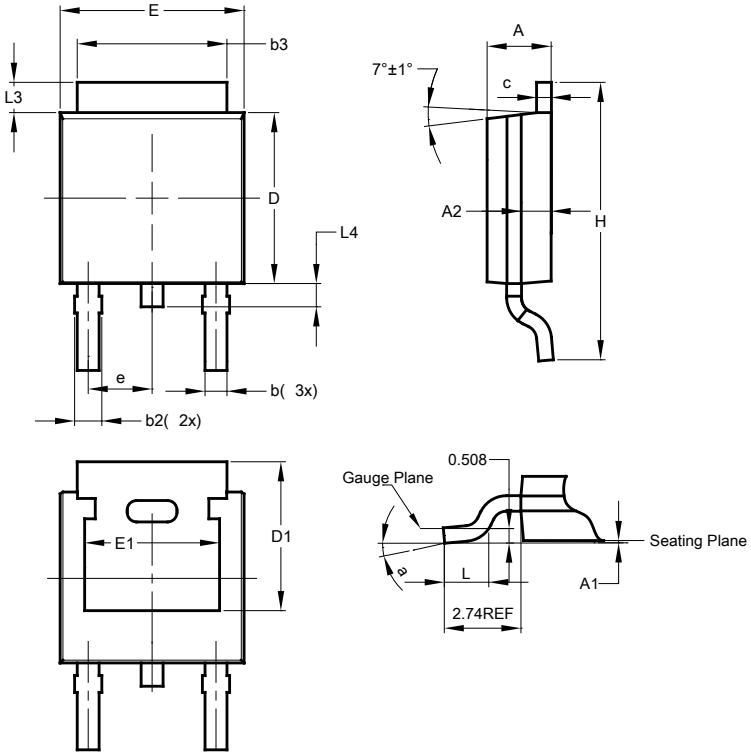
Package Outline Dimensions

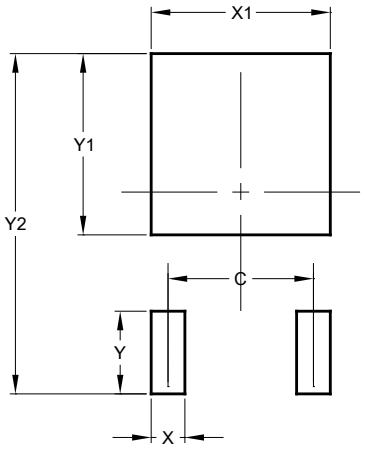
Please see AP02002 at http://www.diodes.com/datasheets/ap02002.pdf for latest version.



TO252 (DPAK)			
Dim	**Min**	**Max**	**Typ**
A	2.19	2.39	2.29
A1	0.00	0.13	0.08
A2	0.97	1.17	1.07
b	0.64	0.88	0.783
b2	0.76	1.14	0.95
b3	5.21	5.46	5.33
c	0.45	0.58	0.531
D	6.00	6.20	6.10
D1	5.21	-	-
e	-	-	2.286
E	6.45	6.70	6.58
E1	4.32	-	-
H	9.40	10.41	9.91
L	1.40	1.78	1.59
L3	0.88	1.27	1.08
L4	0.64	1.02	0.83
a	0°	10°	-
All Dimensions in mm			

Suggested Pad Layout

Please see AP02001 at http://www.diodes.com/datasheets/ap02001.pdf for the latest version.



Dimensions	Value (in mm)
C	4.572
X	1.060
X1	5.632
Y	2.600
Y1	5.700
Y2	10.700

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